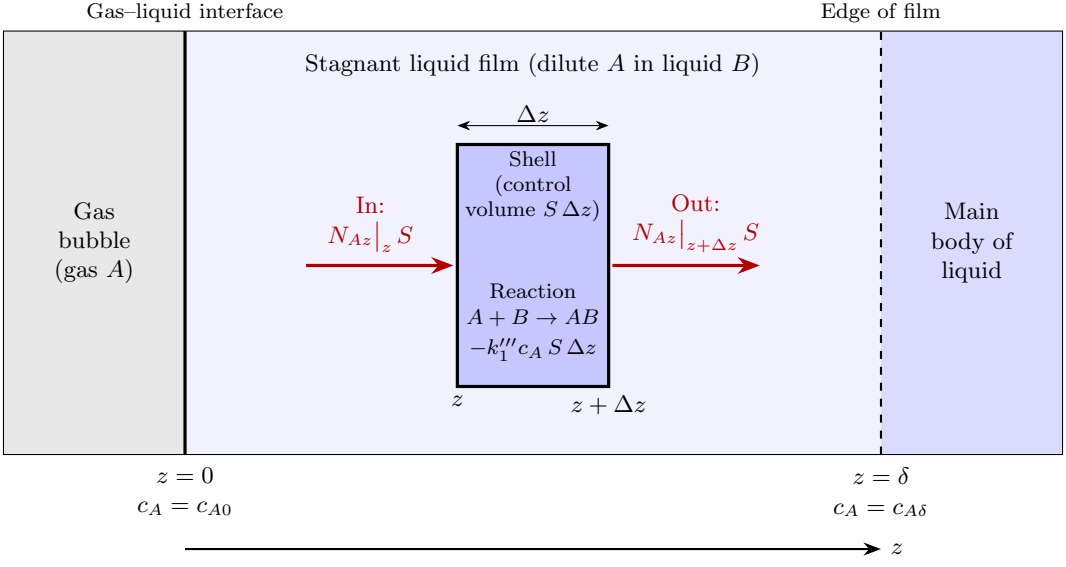

Gas–liquid interface
Edge of film
Stagnant liquid film (dilute A in liquid B)
Δz
Shell
(control
volume $S\,\Delta z$)
In:
$N_{Az}\big|_z\, S$
Out:
$N_{Az}\big|_{z+\Delta z}\, S$
Reaction
$A + B \rightarrow AB$
$-k_1''' c_A\, S\, \Delta z$
Gas
bubble
(gas A)
Main
body of
liquid
z
$z + \Delta z$
$z = 0$
$c_A = c_{A0}$
$z = \delta$
$c_A = c_{A\delta}$
z